

Module 10: Epigenetics — Study Questions

1. What does "gene expression" mean?
2. Every cell in your body has the same DNA. Why do muscle cells and nerve cells look different?
3. What is a transcription factor?
4. What is a promoter? What is an enhancer?
5. What is an operon?
6. How does the lac operon work when lactose is absent?
7. How does the lac operon work when lactose is present?
8. What is the difference between an inducible operon and a repressible operon?
9. What is epigenetics? How is it different from a mutation?
10. What is DNA methylation? Does it turn genes on or off?
11. What are histones? What does histone acetylation do?
12. What is the difference between euchromatin and heterochromatin?
13. What is X-inactivation? What is a Barr body?
14. How do calico cats demonstrate X-inactivation?
15. What is alternative splicing?
16. What is miRNA? What does it do?
17. Give two examples of how the environment can affect gene expression.
18. Why might identical twins develop different traits as they age?

19. A tumor suppressor gene is heavily methylated but not mutated. Could this cause cancer? Explain.
20. Summarize: How do cells with identical DNA become different types of cells?